EBS01003

CHASSIS SPECIFICATIONS

*1 For models equipped with panel wheels and oil damper shock absorbers
*2 For models equipped with cast wheels and oil damper shock absorbers
*3 For models equipped with cast wheels and gas-oil damper shock absorbers
*4 For models equipped with cast wheels, gas-oil damper shock absorbers, and overfenders

Item	Standard	Limit
Steering		
Steering bearing type	Ball and race bearing	----
Front suspension		
Shock absorber stroke	108.0 mm (4.25 in) *1, *2	----
	110.0 mm (4.33 in) *3, *4	----
Spring free length	313.0 mm (12.32 in) *1, *2	----
	336.6 mm (13.25 in) *3, *4	----
Installed length	247.9 mm (9.76 in) *1, *2	----
	260.5 mm (10.26 in) *3, *4	----
Spring rate (K1)	19.40 N/mm (1.98 kgf/mm, 110.77 lb/in)	----
Spring stroke (K1)	0.0 ~ 108.0 mm (0.00 ~ 4.25 in) *1, *2	----
	0.0 ~ 110.0 mm (0.00 ~ 4.33 in) *3, *4	----
Optional spring available	No	----
Rear suspension		
Rear shock absorber assembly	81.0 mm (3.19 in) *1, *2	----
	85.0 mm (3.35 in) *3, *4	----
Spring free length	328.0 mm (12.91 in) *1, *2	----
	286.3 mm (11.27 in) *3, *4	----
Installed length	273.2 mm (10.76 in) *1, *2	----
	235.0 mm (9.25 in) *3, *4	----
Spring rate (K1)	44.10 N/mm (4.50 kgf/mm, 251.81 lb/in)	----
Spring rate (K2)	117.70 N/mm (12.00 kgf/mm, 672.07 lb/in) *1, *2	----
Spring stroke (K1)	0.0 ~ 60.0 mm (0.00 ~ 2.36 in) *1, *2	----
	0.0 ~ 85.0 mm (0.00 ~ 3.35 in) *3, *4	----
Spring stroke (K2)	60.0 ~ 81.0 mm (2.36 ~ 3.19 in) *1, *2	----
Optional spring available	No	----
Front wheel		
Wheel type	Panel wheel *1	----
	Cast wheel *2, *3, *4	----
Rim size	12 × 6.0 AT	----
Rim material	Steel *1	----
	Aluminum *2, *3, *4	----
Max. radial wheel runout	----	2.0 mm (0.08 in)
Max. lateral wheel runout	----	2.0 mm (0.08 in)

Item	Standard	Limit
Rear wheel		
Wheel type	Panel wheel *1	----
	Cast wheel *2, *3, *4	----
Rim size	12 × 7.5 AT	----
Rim material	Steel *1	----
	Aluminum *2, *3, *4	----
Max. radial wheel runout	----	2.0 mm (0.08 in)
Max. lateral wheel runout	----	2.0 mm (0.08 in)
Front disc brake		
Type	Dual	----
Disc outside diameter × thickness	200.0 × 3.5 mm (7.87 × 0.14 in)	----
Brake disc minimum thickness	3.0 mm (0.12 in)	----
Brake disc maximum deflection	0.1 mm (0.004 in)	----
Pad thickness inner	5.2 mm (0.20 in)	1.5 mm (0.06 in)
Pad thickness outer	5.2 mm (0.20 in)	1.5 mm (0.06 in)
Master cylinder inside diameter	17.46 mm (0.69 in)	----
Caliper cylinder inside diameter	27.00 mm (1.06 in)	----
Recommended fluid	DOT 4	----
Rear disc brake		
Type	Dual	----
Disc outside diameter × thickness	184.6 × 3.5 mm (7.27 × 0.14 in)	----
Brake disc minimum thickness	3.0 mm (0.12 in)	----
Brake disc maximum deflection	0.1 mm (0.004 in)	----
Pad thickness inner	5.2 mm (0.20 in)	1.5 mm (0.06 in)
Pad thickness outer	5.2 mm (0.20 in)	1.5 mm (0.06 in)
Caliper cylinder inside diameter	25.40 mm (1.00 in)	----
Recommended fluid	DOT 4	----

Item	Standard	Limit
Parking brake		
Type	Disc	----
Disc outside diameter × thickness	165.0 × 3.5 mm (6.50 × 0.14 in)	----
Brake disc minimum thickness	3.0 mm (0.12 in)	----
Brake disc maximum deflection	0.1 mm (0.004 in)	----
Brake pad lining thickness-inner	3.2 mm (0.13 in)	1.0 mm (0.04 in)
Brake pad lining thickness-outer	3.2 mm (0.13 in)	1.0 mm (0.04 in)
Brake lever and brake pedal		
Accelerator pedal free play	0 mm (0 in)	----
Brake pedal free play	0 mm (0 in)	----
Parking brake cable free play	The maximum free play is equal to one click of the parking brake lever.	----

EBS01004

ELECTRICAL SPECIFICATIONS

Item	Standard	Limit
System voltage	12 V	----
Ignition system		
Ignition timing (B.T.D.C.)	12°/1,400 r/min	----
Advancer type	Digital	----
Transistorized coil ignition		
Crankshaft position sensor resistance/color	459 ~ 561 Ω at 20 °C (68 °F)/ gray/black	----
ECU (engine control unit)		
Model/manufacturer	F8T83871/MITSUBISHI	----
Ignition coil		
Model/manufacturer	2JN/YAMAHA	----
Minimum ignition spark gap	6.0 mm (0.24 in)	----
Primary coil resistance	2.16 ~ 2.64 Ω at 20 °C (68 °F)	----
Secondary coil resistance	8.64 ~ 12.96 kΩ at 20 °C (68 °F)	----
Spark plug cap		
Material	Resin	----
Resistance	10.0 kΩ	----
AC magneto		
Model/manufacturer	F004T39372/MITSUBISHI	----
Standard output	14.0 V 33.5 A at 5,000 r/min	----
Stator coil resistance/color	0.099 ~ 0.121 Ω at 20 °C (68 °F)/ white-white	----
Rectifier/regulator		
Type	Semiconductor-short-circuit	----
Model/manufacturer	FH012AA/SHINDENGEN	----
No load regulated voltage (DC)	14.2 ~ 14.8 V	----
Rectifier capacity	50.0 A	----
Withstand voltage	40.0 V	----
Electric starting system		
Type	Constant mesh	----
Starter motor		
Model/manufacturer	SM-13/MITSUBA	----
Power output	0.80 kW	----
Armature coil resistance	0.0250 ~ 0.0350 Ω at 20 °C (68 °F)	----
Brush overall length	12.5 mm (0.49 in)	5.00 mm (0.20 in)
Spring force	7.65 ~ 10.01 N (780 ~ 1,021 gf, 27.54 ~ 36.03 oz)	----
Commutator diameter	28.0 mm (1.10 in)	27.0 mm (1.06 in)
Mica undercut (depth)	0.70 mm (0.03 in)	----

Item	Standard	Limit
Starter relay		
Model/manufacturer	RC19-080A/MITSUBA	----
Amperage rating	180.0 A	----
Coil winding resistance	4.18 ~ 4.62 Ω at 20 °C (68 °F)	----
Fuel sender		
Sender unit resistance (full)	19.00 ~ 21.00 Ω	----
Sender unit resistance (empty)	137.00 ~ 143.00 Ω	----
Fuel injection system relay		
Model/manufacturer	ACM33211/MATSUSHITA	----
Coil resistance	96.0 Ω	----
Radiator fan motor relay		
Model/manufacturer	ACM33211/MATSUSHITA	----
Coil resistance	96.0 Ω	----
Load control relay		
Model/manufacturer	ACM33211/MATSUSHITA	----
Coil resistance	96.0 Ω	----
Four-wheel-drive motor relay 3		
Model/manufacturer	ACM33211/MATSUSHITA	----
Coil resistance	96.0 Ω	----
Headlight relay		
Model/manufacturer	G8HN-1C4T-DJ/OMRON	----
Coil resistance	105.0 Ω	----
Four-wheel-drive motor relay 1		
Model/manufacturer	G8HN-1C4T-DJ/OMRON	----
Coil resistance	105.0 Ω	----
Four-wheel-drive motor relay 2		
Model/manufacturer	G8HN-1C4T-DJ/OMRON	----
Coil resistance	105.0 Ω	----
Circuit breaker		
Circuit breaker type	Fuse	----
Fuses		
Main fuse	40.0 A	----
Headlight fuse	15.0 A	----
Signaling system fuse	10.0 A	----
Ignition fuse	10.0 A	----
Auxiliary DC jack fuse	10.0 A	----
Fuel injection system fuse	10.0 A	----
Four-wheel-drive motor fuse	10.0 A	----
Backup fuse	10.0 A	----
Radiator fan motor fuse	25.0 A	----
Spare fuse	25.0 A	----
	15.0 A	----
	10.0 A	----

EBS01005

TIGHTENING TORQUES

ENGINE TIGHTENING TORQUES

Item	Part name	Thread size	Q'ty	Tightening torque			Remarks
				Nm	m · kg	ft · lb	
Cylinder head (exhaust pipe)	Stud bolt	M8	4	15	1.5	11	
Cylinder head	Bolt	M9	4	35	3.5	25	M
Cylinder head	Bolt	M9	2	38	3.8	27	E
Cylinder head	Bolt	M6	2	10	1.0	7.2	
Spark plug	—	M10	1	13	1.3	9.4	
Oil check bolt	Bolt	M8	1	10	1.0	7.2	
Cylinder	Bolt	M10	4	50	5.0	36	See TIP. E
AC magneto rotor	Nut	M16	1	60	6.0	43	E
Balancer driven gear	Nut	M18	1	80	8.0	58	Use a lock washer. E
Thermostat cover	Bolt	M6	2	10	1.0	7.2	
Cylinder head air bleed bolt	Bolt	M6	1	10	1.0	7.2	
Valve adjusting screw	Nut	M6	4	14	1.4	10	
Decompression assembly	Bolt	M7	2	20	2.0	14	
Timing chain guide (intake side)	Bolt	M6	2	10	1.0	7.2	LT
Timing chain tensioner cap	Bolt	M16	1	20	2.0	14	
Timing chain tensioner	Bolt	M6	2	10	1.0	7.2	
Bearing retainer (camshaft)	Bolt	M6	2	10	1.0	7.2	LT
Camshaft sprocket cover	Bolt	M6	2	10	1.0	7.2	
Tappet cover	Bolt	M6	8	10	1.0	7.2	
Camshaft sprocket	Bolt	M7	2	20	2.0	14	
Crankcase	Bolt	M8	3	26	2.6	19	
	Bolt	M6	4	10	1.0	7.2	
	Bolt	M6	8	10	1.0	7.2	
Engine oil drain bolt	Bolt	M14	1	30	3.0	22	
Oil filter cartridge	—	M20	1	17	1.7	12	
Oil filter cartridge union bolt	Union bolt	M20	1	68	6.8	49	E
Oil pipe (dipstick)	Bolt	M6	1	10	1.0	7.2	
Oil delivery pipe 1	Union bolt	M14	2	35	3.5	25	
Oil delivery pipe 1	Union bolt	M10	1	20	2.0	14	
Oil delivery pipe 1	Bolt	M6	1	10	1.0	7.2	
Oil delivery pipe 2	Bolt	M6	2	10	1.0	7.2	
Oil delivery pipe 3	Bolt	M6	2	10	1.0	7.2	
Oil pump	Bolt	M6	3	10	1.0	7.2	
Oil pump driven gear	Nut	M10	1	22	2.2	16	
Timing chain guide	Bolt	M6	2	10	1.0	7.2	LT
V-belt cooling duct 1	Bolt	M6	2	7	0.7	5.1	
Air filter case	Bolt	M6	4	7	0.7	5.1	

Item	Part name	Thread size	Q'ty	Tightening torque			Remarks
				Nm	m · kg	ft · lb	
Air intake duct	Bolt	M6	2	7	0.7	5.1	
Muffler	Bolt	M8	2	20	2.0	14	
Muffler bracket	Bolt	M8	4	30	3.0	22	
Exhaust pipe	Nut	M8	4	20	2.0	14	
Tailpipe	Bolt	M6	3	10	1.0	7.2	
Radiator inlet pipe holder	Bolt	M6	1	7	0.7	5.1	
Radiator outlet pipe holder	Bolt	M6	1	7	0.7	5.1	
Oil cooler inlet pipe 1/oil cooler outlet pipe 1 holder	Bolt	M6	1	7	0.7	5.1	
Oil cooler inlet pipe 2/oil cooler outlet pipe 2 holder	Bolt	M6	2	7	0.7	5.1	
Oil cooler inlet pipe 1/oil cooler outlet pipe 1 and oil cooler	Bolt	M17	2	21	2.1	15	
Oil cooler	Nut	M6	4	7	0.7	5.1	
Radiator	Bolt	M6	2	7	0.7	5.1	
Coolant reservoir	Bolt	M6	2	7	0.7	5.1	
Water pump housing	Bolt	M6	2	10	1.0	7.2	
Coolant drain bolt	Bolt	M6	1	10	1.0	7.2	
Water pump air bleed bolt	Bolt	M6	1	10	1.0	7.2	
Water pump outlet pipe	Bolt	M6	1	10	1.0	7.2	
Water jacket joint	Bolt	M6	2	10	1.0	7.2	
Crankshaft end accessing screw	Bolt	M36	1	10	1.0	7.2	
Timing mark accessing screw	Bolt	M14	1	6	0.6	4.3	
Drive belt cover	Bolt	M6	12	10	1.0	7.2	
Drive belt case	Bolt	M6	8	10	1.0	7.2	
V-belt cooling duct 1 bracket	Bolt	M6	2	10	1.0	7.2	
Bearing housing (primary sheave assembly)	Bolt	M6	4	10	1.0	7.2	
Bearing retainer (bearing housing)	Bolt	M6	1	10	1.0	7.2	
AC magneto cover	Bolt	M6	11	10	1.0	7.2	
AC magneto/crankshaft position sensor lead holder	Bolt	M5	2	7	0.7	5.1	
Starter one-way clutch	Screw	M8	3	30	3.0	22	
Clutch housing assembly	Bolt	M6	9	10	1.0	7.2	
Clutch carrier assembly	Nut	M22	1	190	19.0	140	Stake. 
Middle drive pinion gear nut	Nut	M22	1	190	19.0	140	Stake.
Middle driven shaft bearing housing	Bolt	M8	4	32	3.2	23	
Middle drive shaft bearing retainer	Screw	M8	4	29	2.9	21	Stake. 
Middle driven shaft bearing retainer	Nut	M55	1	80	8.0	58	Left-hand thread 